

Multivariate Real Analysis

Muchang Bahng

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In here, we extend the results of univariate real analysis to multivariate and/or vector-valued functions. In practice, multivariate calculus is used, and there are many new results that arise in the multivariate case. The case for computing limits and continuity on multivariate functions is very straightforward, since these are topological properties. Generally, to prove that a limit is something, we just use ϵ - δ , and to actually *compute* what the limit might be, we can just compute $f(a)$ assuming that it is continuous, or take some sort of path p to take the univariate limit. We can also show that the multivariate limit doesn't exist by taking two sequences or two path functions where the limits do not equal each other.

However, the definition of the derivative and the integral will need to be generalized. Note that both the domain and codomain of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are vector spaces. However, Euclidean spaces have a lot of structure on them, and it is nice to identify the essential properties we need from these sets. It is essential that we work in vector spaces because when we define a derivative, we usually see some form that looks like

$$\frac{f(x+h) - f(x)}{h} \quad (1)$$

Note the operations used here. First, we want a notion of addition in the domain ($x+h$) and the codomain $f(x+h) - f(x)$, along with some scalar multiplication when we multiply by $1/h$. A vector space precisely supports these operations and therefore is a natural choice. It is immediate that to define convergence, we definitely need a topology. We will see later that we want to define multivariate derivatives by adding a norm to this term, requiring the use of a normed vector space. Completion is clearly essential as we have seen in single-variable analysis.

Definition 0.1 (Banach Space)
A Banach space is a normed complete vector space.

Note that by extending the dimension, we have essentially lost the ordering $\leq$ on these spaces, along with the field properties. Therefore, we will need to adapt our definitions accordingly.

Differentiation

1.1 Partial, Directional, and Path Derivatives

A simple way to extract derivatives is to fix all of the input components except for one, and treat f as a single-variable function. This results in—for now—a completely separate notion of a derivative.

Definition 1.1 (Partial Derivative)

Given function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the i th **partial derivative** of f at $a \in E$ is defined

$$\left. \frac{\partial f}{\partial x_i} \right|_a := \lim_{h \rightarrow 0} \frac{f(a + he_i) - f(a)}{h} \quad (2)$$

where e_i is the i th basis vector of $\mathbb{R}^n$.

Since we are fixing every other parameter except for the i th one, we can find the partial derivative by differentiating the function w.r.t. x_i and fixing all other variables.

Example 1.1 (Partial Derivatives of Saddle Points)

To find all partial derivatives of the function

$$f(x, y) = x^2 - y^2 \quad (3)$$

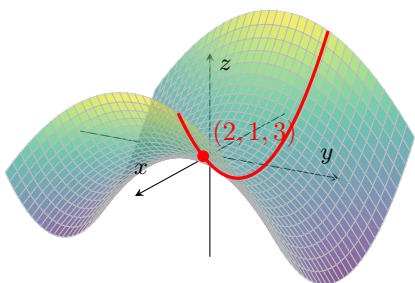
We compute

$$\frac{\partial f}{\partial x} = 2x \quad (4)$$

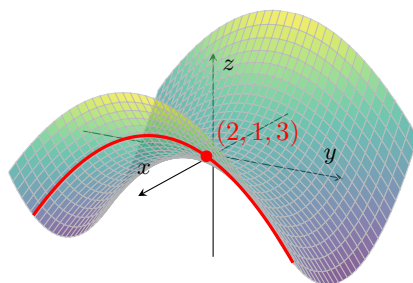
$$\frac{\partial f}{\partial y} = -2y. \quad (5)$$

At the point $(2, 1)$, these become

$$\frac{\partial f}{\partial x}(2, 1) = 4 \quad \text{and} \quad \frac{\partial f}{\partial y}(2, 1) = -2. \quad (6)$$



(a) Slicing the saddle surface along the plane $y = 1$, the resulting curve has slope 4 at $(2, 1, 3)$.



(b) Slicing along the plane $x = 2$ gives a curve with slope -2 at the same point.

Figure 1: Two cross-sections of the saddle surface $z = x^2 - y^2$ through the point $(2, 1, 3)$. Each slice isolates one partial derivative by holding the other variable fixed.

Example 1.2 (Partial Derivatives Using Differentiation Rules)

To find all partial derivatives of the function

$$f(x, y, z) = x \ln(y^2 + z^2), \quad (7)$$

We compute

$$\frac{\partial f}{\partial x} = \ln(y^2 + z^2) \quad (8)$$

$$\frac{\partial f}{\partial y} = x \cdot \frac{1}{y^2 + z^2} \cdot (2y) = \frac{2xy}{y^2 + z^2} \quad (9)$$

$$\frac{\partial f}{\partial z} = x \cdot \frac{1}{y^2 + z^2} \cdot (2z) = \frac{2xz}{y^2 + z^2} \quad (10)$$

Example 1.3 (Partial Derivatives Using Limit Definition)

Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0). \end{cases} \quad (11)$$

We compute the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ using the limit definition.

First, for the partial derivative with respect to x , we have

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} \quad (12)$$

$$= \lim_{h \rightarrow 0} \frac{\frac{h(0)}{h^2 + 0^2} - 0}{h} \quad (13)$$

$$= \lim_{h \rightarrow 0} \frac{0}{h} \quad (14)$$

$$= 0. \quad (15)$$

Similarly, for the partial derivative with respect to y , we have

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} \quad (16)$$

$$= \lim_{h \rightarrow 0} \frac{\frac{0(h)}{0^2 + h^2} - 0}{h} \quad (17)$$

$$= \lim_{h \rightarrow 0} \frac{0}{h} \quad (18)$$

$$= 0. \quad (19)$$

Therefore,

$$f_x(0, 0) = 0 \quad \text{and} \quad f_y(0, 0) = 0. \quad (20)$$

Note that even though the limit of $f(x, y)$ as $(x, y) \rightarrow (0, 0)$ does not exist, the partial derivatives at $(0, 0)$ still exist.

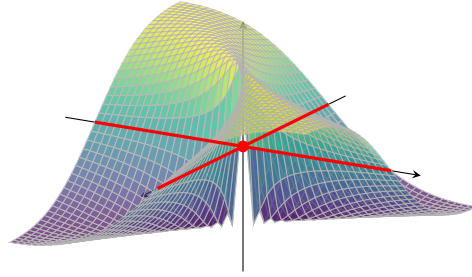


Figure 2: Both coordinate cross-sections of $f(x, y) = \frac{xy}{x^2 + y^2}$ through the origin lie on the line $z = 0$, which shows why $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$ even though the full two-variable limit does not exist.

The partial derivative looks at the function as it is approaching a along an axis, but we can consider any direction in the domain.

Definition 1.2 (Directional Derivative)

Given function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the **directional derivative** of f at $a \in E$ in direction $u \in \mathbb{R}^n$ is defined

$$D_v f_a := \lim_{h \rightarrow 0} \frac{f(a + hu) - f(a)}{h} \quad (21)$$

It should be obvious that if all directional derivatives exist, then the partials exist (since we can just set the directional vectors to be the unit vectors).

When computing directional derivatives, it is convenient to normalize the directional vector v to be unit length so that it coincides with the partial derivatives. We don't technically need to set $\|v\| = 1$, but if we have two vectors v and a scaled cv , then the directional derivatives will also be scaled ($\nabla_{cv} f(a) = c \nabla_v f(a)$), so we will only work with unit directional vectors. Some say that this restriction is undesirable, since it loses the linearity of the function $v \mapsto \partial_v f(a)$.

Example 1.4

Let

$$f(x, y) = x^2 y + 4y^2. \quad (22)$$

We compute the directional derivative of f at the point $(2, 1)$ in the direction of the vector

$$v = (1, 1).$$

First, we normalize v . Since

$$\|v\| = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad (23)$$

the unit direction vector is

$$u = \frac{v}{\|v\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right). \quad (24)$$

By the limit definition of the directional derivative,

$$D_u f(2, 1) = \lim_{h \rightarrow 0} \frac{f\left(\left(2, 1\right) + hu\right) - f(2, 1)}{h} \quad (25)$$

$$= \lim_{h \rightarrow 0} \frac{f\left(2 + \frac{h}{\sqrt{2}}, 1 + \frac{h}{\sqrt{2}}\right) - f(2, 1)}{h}. \quad (26)$$

Now,

$$f(2, 1) = 2^2(1) + 4(1)^2 = 8. \quad (27)$$

Also,

$$f\left(2 + \frac{h}{\sqrt{2}}, 1 + \frac{h}{\sqrt{2}}\right) = \left(2 + \frac{h}{\sqrt{2}}\right)^2 \left(1 + \frac{h}{\sqrt{2}}\right) + 4\left(1 + \frac{h}{\sqrt{2}}\right)^2 \quad (28)$$

$$= 8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}}. \quad (29)$$

Therefore,

$$D_u f(2, 1) = \lim_{h \rightarrow 0} \frac{8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}} - 8}{h} \quad (30)$$

$$= \lim_{h \rightarrow 0} \left(\frac{16}{\sqrt{2}} + \frac{9h}{2} + \frac{h^2}{2\sqrt{2}} \right) \quad (31)$$

$$= \frac{16}{\sqrt{2}} \quad (32)$$

$$= 8\sqrt{2}. \quad (33)$$

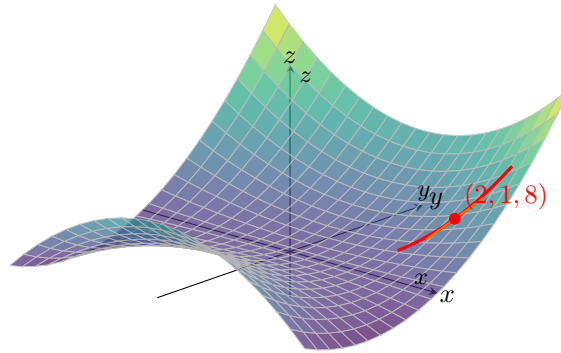


Figure 3: The surface $z = x^2y + 4y^2$ together with the red curve obtained by moving from $(2, 1)$ in the direction $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. The orange line is the tangent line, whose slope is the directional derivative $D_u f(2, 1) = 8\sqrt{2}$.

Example 1.5

Let

$$f(x, y, z, w) = x^2y + z \cos(w). \quad (34)$$

We compute the directional derivative of f at the point $(1, 2, 0, 0)$ in the direction of the vector

$$v = (1, 1, 1, 1).$$

First, we normalize v . Since

$$\|v\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2, \quad (35)$$

the unit direction vector is

$$u = \frac{v}{\|v\|} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \quad (36)$$

By the limit definition of the directional derivative,

$$D_u f(1, 2, 0, 0) = \lim_{h \rightarrow 0} \frac{f((1, 2, 0, 0) + hu) - f(1, 2, 0, 0)}{h} \quad (37)$$

$$= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) - f(1, 2, 0, 0)}{h}. \quad (38)$$

Now,

$$f(1, 2, 0, 0) = 1^2(2) + 0 \cos(0) = 2. \quad (39)$$

Also,

$$f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) = \left(1 + \frac{h}{2}\right)^2 \left(2 + \frac{h}{2}\right) + \frac{h}{2} \cos\left(\frac{h}{2}\right) \quad (40)$$

$$= \left(1 + h + \frac{h^2}{4}\right) \left(2 + \frac{h}{2}\right) + \frac{h}{2} \cos\left(\frac{h}{2}\right) \quad (41)$$

$$= 2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2} \cos\left(\frac{h}{2}\right). \quad (42)$$

Therefore,

$$D_u f(1, 2, 0, 0) = \lim_{h \rightarrow 0} \frac{2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2} \cos\left(\frac{h}{2}\right) - 2}{h} \quad (43)$$

$$= \lim_{h \rightarrow 0} \left(\frac{5}{2} + h + \frac{h^2}{8} + \frac{1}{2} \cos\left(\frac{h}{2}\right) \right) \quad (44)$$

$$= \frac{5}{2} + \frac{1}{2} \quad (45)$$

$$= 3. \quad (46)$$

But be careful! The existence of partials does not imply the existence of all directional derivatives!

Example 1.6 (Existence of Partial $\not\Rightarrow$ Existence of Directional Derivatives)

Consider the function

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

The partial derivatives exist everywhere. Away from the origin we can simply compute

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= \frac{x_2(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_1}{(x_1^2 + x_2^2)^2} = \frac{-x_1^2 x_2 + x_2^3}{(x_1^2 + x_2^2)^2} \\ \frac{\partial f}{\partial x_2} &= \frac{x_1(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_2}{(x_1^2 + x_2^2)^2} = \frac{x_1^3 - x_1 x_2^2}{(x_1^2 + x_2^2)^2} \end{aligned}$$

As for the partials at the origin, we must compute using the limit rule.

$$\begin{aligned} \partial_{x_1} f(0) &= \lim_{h \rightarrow 0} \frac{f(0 + he_1) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h, 0)}{h} = 0 \\ \partial_{x_2} f(0) &= \lim_{h \rightarrow 0} \frac{f(0 + he_2) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(0, h)}{h} = 0 \end{aligned}$$

However, the directional derivative taken in direction $v = (1, 1)$ gives

$$\begin{aligned}\nabla_{(1,1)}f(0,0) &= \lim_{h \rightarrow 0} \frac{f(0+h(1,1)) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h,h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{2h}\end{aligned}$$

which does not have a limit as $h \rightarrow 0$. To visualize this, let's look at the values of f along various lines in $\mathbb{R}^2$.

1. $f = 0$ at the line $x_1 = 0$ and $x_2 = 0$, which is why the partials are 0.
2. $f = \frac{1}{2}$ at the line where $x_1 = x_2$, except for the point $(0,0)$, where $f = 0$, which is why the limit in the direction doesn't exist.

Definition 1.3 (Path Derivative)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, $a \in E$, and $p : \mathbb{R} \rightarrow \mathbb{R}^n$ be a path function where $p(t_0) = a$. Then, the **path derivative** of f at a with respect to path p is defined

$$D_p f(a) := \lim_{t \rightarrow t_0} \frac{f(p(t)) - f(p(t_0))}{h} \quad (47)$$

Example 1.7

Let

$$f(x, y) = x^2 + y, \quad (48)$$

and let

$$p(t) = (1 + t, 1 + t^2). \quad (49)$$

Since

$$p(0) = (1, 1), \quad (50)$$

we compute the path derivative of f at $a = (1, 1)$ with respect to the path p .

By the definition of the path derivative,

$$D_p f(1, 1) = \lim_{t \rightarrow 0} \frac{f(p(t)) - f(p(0))}{t - 0} \quad (51)$$

$$= \lim_{t \rightarrow 0} \frac{f(1 + t, 1 + t^2) - f(1, 1)}{t}. \quad (52)$$

Now,

$$f(1, 1) = 1^2 + 1 = 2, \quad (53)$$

and

$$f(1 + t, 1 + t^2) = (1 + t)^2 + (1 + t^2) \quad (54)$$

$$= 1 + 2t + t^2 + 1 + t^2 \quad (55)$$

$$= 2 + 2t + 2t^2. \quad (56)$$

Therefore,

$$D_p f(1, 1) = \lim_{t \rightarrow 0} \frac{2 + 2t + 2t^2 - 2}{t} \quad (57)$$

$$= \lim_{t \rightarrow 0} \frac{2t + 2t^2}{t} \quad (58)$$

$$= \lim_{t \rightarrow 0} (2 + 2t) \quad (59)$$

$$= 2. \quad (60)$$

Example 1.8

Let

$$f(x, y) = (xy, x^2 - y), \quad (61)$$

and let

$$p(t) = (\cos t, \sin t). \quad (62)$$

Since

$$p(0) = (1, 0), \quad (63)$$

we compute the path derivative of f at $a = (1, 0)$ with respect to the path p .

By the definition of the path derivative,

$$D_p f(1, 0) = \lim_{t \rightarrow 0} \frac{f(p(t)) - f(p(0))}{t - 0} \quad (64)$$

$$= \lim_{t \rightarrow 0} \frac{f(\cos t, \sin t) - f(1, 0)}{t}. \quad (65)$$

Now,

$$f(1, 0) = (0, 1), \quad (66)$$

and

$$f(\cos t, \sin t) = (\cos t \sin t, \cos^2 t - \sin t). \quad (67)$$

Therefore,

$$D_p f(1, 0) = \lim_{t \rightarrow 0} \frac{(\cos t \sin t, \cos^2 t - \sin t) - (0, 1)}{t} \quad (68)$$

$$= \lim_{t \rightarrow 0} \left(\frac{\cos t \sin t}{t}, \frac{\cos^2 t - \sin t - 1}{t} \right). \quad (69)$$

We compute each component separately. For the first component,

$$\lim_{t \rightarrow 0} \frac{\cos t \sin t}{t} = \lim_{t \rightarrow 0} \cos t \cdot \frac{\sin t}{t} \quad (70)$$

$$= 1. \quad (71)$$

For the second component,

$$\lim_{t \rightarrow 0} \frac{\cos^2 t - \sin t - 1}{t} = \lim_{t \rightarrow 0} \left(\frac{\cos^2 t - 1}{t} - \frac{\sin t}{t} \right) \quad (72)$$

$$= \lim_{t \rightarrow 0} \left(\frac{-\sin^2 t}{t} - \frac{\sin t}{t} \right) \quad (73)$$

$$= 0 - 1 \quad (74)$$

$$= -1. \quad (75)$$

Hence,

$$D_p f(1, 0) = (1, -1). \quad (76)$$

1.2 Total Derivatives

Definition 1.4 (Total/Frechet Derivative)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $a \in E$. f is **differentiable** at a if it satisfies either of the equivalent conditions.

1. *Linear Approximation*. There exists a linear map $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} = 0 \quad (77)$$

2. *Fundamental Increment Lemma*. There exists a linear map $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and an infinitesimal^b function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined for sufficiently small nonzero $h \in \mathbb{R}^n$ such that

$$f(a+h) = f(a) + Df_a h + \Phi(h)\|h\|, \quad \lim_{h \rightarrow 0} \Phi(h) = 0 \quad (78)$$

If such a map Df_a exists, then it is unique, and we call this the **total derivative**—or **Frechet derivative**—of f at a .

^aNote that when $m = 1$, then Df_a is a linear functional, i.e. a dual vector.

^bAs in $\Phi(h) \rightarrow 0$ as $h \rightarrow 0$.

Proof. We'll first prove that the two are equivalent.

1. ($\rightarrow$). Given that linear approximation holds, we can construct the map

$$\Phi(h) := \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \quad (79)$$

Therefore, by our assumption and absolute homogeneity of the norm, we have

$$0 = \lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \quad (80)$$

$$= \lim_{h \rightarrow 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \quad (81)$$

$$= \lim_{h \rightarrow 0} \|\Phi(h)\| \quad (82)$$

Since the norm $v \mapsto \|v\|$ is continuous, this implies that

$$\lim_{h \rightarrow 0} \Phi(h) = 0 \quad (83)$$

2. ($\leftarrow$). Given that the fundamental increment lemma holds, we can see that for nonzero h ,

$$\Phi(h) = \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \quad (84)$$

Therefore, by taking the limit of both sides, we must have

$$0 = \lim_{h \rightarrow 0} \Phi(h) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \quad (85)$$

Therefore, by continuity of the norm and absolute homogeneity,

$$0 = \|0\| = \left\| \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \quad (86)$$

$$= \lim_{h \rightarrow 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \quad (87)$$

$$= \lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \quad (88)$$

and we are done.

Finally, to prove uniqueness of the derivative, it seems more convenient to use the fundamental increment lemma. Say that there are two derivatives, A_1 and A_2 , each with their own infinitesimal functions Φ_1, Φ_2 defined in the intersection of the open neighborhoods of a , which is also an open neighborhood. Then,

$$f(a+h) = f(a) + A_1 h + \Phi_1(h)\|h\| \quad (89)$$

$$f(a+h) = f(a) + A_2 h + \Phi_2(h)\|h\| \quad (90)$$

So, by subtracting them and rearranging, we have

$$0 = (A_1 - A_2)h + (\Phi_1 - \Phi_2)(h) \quad (91)$$

So for $h \neq 0$, we have

$$(A_2 - A_1) \frac{h}{\|h\|} = (\Phi_1 - \Phi_2)(h) \quad (92)$$

The left hand side is a linear map of h and is in fact a constant c since $\frac{h}{\|h\|}$ is a unit vector. So by taking the limits of both sides, we get

$$c = \lim_{h \rightarrow 0} (A_2 - A_1) \frac{h}{\|h\|} = \lim_{h \rightarrow 0} (\Phi_1 - \Phi_2)(h) = 0 \quad (93)$$

Therefore, $c = 0 \implies A_2 - A_1 = 0$. So $A_1 = A_2$.

The reason we want E to be open is that we require that $x+h$ to also be in D for sufficiently small h , and we can guarantee this since an ϵ -ball around x is guaranteed to be in E . As in univariate analysis, we first establish the relation to continuity.

Theorem 1.1 (Differentiability Implies Continuity)

If f is differentiable at x , then it is continuous at x .

Proof. By assumption, we have

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Df_x h\|}{\|h\|} = 0 \implies \lim_{h \rightarrow 0} \|f(x+h) - f(x) - Df_x h\| = 0 \quad (94)$$

since the norm is a continuous mapping. But $\lim_{h \rightarrow 0} Df_x h = 0$ since Df_x is linear, and so

$$\lim_{h \rightarrow 0} \|f(x+h) - f(x)\| = 0 \quad (95)$$

Therefore, we have 4 separate notions of derivatives. A natural question to ask is whether the existence of one derivative implies the existence of another.

Theorem 1.2 (Differentiability $\implies$ Existence of All Directionals)

If $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $a \in E$, then all of its directional derivatives exist. Furthermore, given direction vector $u \in \mathbb{R}^n$,

$$D_u f_a = Df_a(u) \quad (96)$$

That is, we can compute the directional derivative by plugging in the direction vector into the linear map Df_a .

Proof. Since f is differentiable at a , the limit

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - Df_a(h)\|}{\|h\|} = 0 \quad (97)$$

for all paths through 0. Therefore, let us choose the path

$$\lim_{h \rightarrow 0} \frac{\|f(a+hu) - f(a) - Df_a(hu)\|}{\|hu\|} = \lim_{h \rightarrow 0} \left\| \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \right\| = 0 \in \mathbb{R} \quad (98)$$

Since the norm is a continuous mapping, this means that

$$0 = \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \quad (99)$$

$$= \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a) - hDf_a(u)}{h} \quad (100)$$

$$= \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a)}{h} - Df_a(u) \quad (101)$$

Therefore,

$$Df_a(u) = \lim_{h \rightarrow 0} \frac{f(a+hu) - f(a)}{h} = D_u f_a \quad (102)$$

Therefore, we know that differentiability (i.e. existence of the total derivative) is the strongest. If this is the case, then the partial and directional derivatives exist, and we also know how the derivative acts on each basis vector. Therefore, we can immediately know that

Lemma 1.3 (Action of Derivative on Basis Vectors)

Let $(e_i)_i$ and (e'_j) be the standard basis vectors of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at a , then the partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist, and

$$Df_a(e_i) = \sum_{j=1}^m \left(\frac{\partial f_j}{\partial x_i}(a) \right) e'_j \quad (103)$$

Proof. The partial derivatives existing is an immediate consequence of the previous theorem. As for the identity, we can see that

$$Df_a(e_i) = D_{e_i} f_a = D_{e_i} \left(\sum_{j=1}^m f_j e'_j \right)_a \quad (104)$$

$$= \sum_{j=1}^m D_{e_i} (f_j e'_j)_a \quad (105)$$

$$= \sum_{j=1}^m D_{e_i} (f_j)_a e'_j \quad (106)$$

$$= \sum_{j=1}^m \frac{\partial f_j}{\partial x_i}(a) e'_j \quad (107)$$

Since our vector spaces come with a basis, with this theorem, we can realize Df_x in a matrix form.

Definition 1.5 (Jacobian)

Given $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ differentiable at x , the **Jacobian** of f at x is the matrix realization of the total derivative, with entries being precisely the partial derivatives.

$$Jf_a = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \quad (108)$$

Therefore, we can see that the derivative maps the i th basis vector $e_i \in \mathbb{R}^n$ to the i th column of our Jacobian matrix.

Analogous to the univariate case, a nice way to visualize the derivative is by looking at the tangent *plane*. In general, we have an *affine hyperplane*. Define this here. Unfortunately, the only types of functions that we can meaningfully visualize are those mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}$.

Definition 1.6 (Tangent Hyperplane)

If $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $a \in \mathbb{R}^n$, then the affine hyperplane tangent to the graph of f at $(a, f(a))$ is the set

$$P = \{(x, f(x)) : z = f(a) + Df_a(x - a)\}. \quad (109)$$

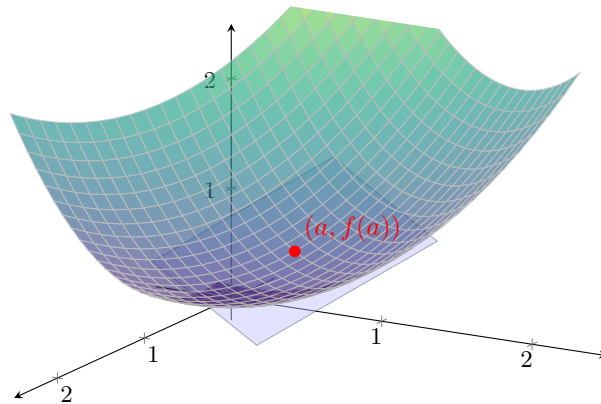


Figure 4: The tangent plane to the surface $z = f(x, y)$ at the point $(a, f(a))$. Here $f(x, y) = \frac{x^2 + y^2}{2}$ and $a = (1, 1)$, so the tangent plane is $z = x + y - 1$.

So, to prove that a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at a , and if so, what its total derivative is, there are essentially two steps.

1. We find a candidate for M by evaluating the partials.

$$M = (\partial_{x_1} f(a) \quad \partial_{x_2} f(a) \quad \cdots \quad \partial_{x_n} f(a)) \quad (110)$$

2. We check to see if the limit is true.

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a) - Mh}{\|h\|} = 0 \quad (111)$$

If it is, then $Df_a = M$, and the "tangent plane" of f at a is defined by the equation $y = f(a) + Mh$.

Example 1.9 (Computing Total Derivative)

The function $f(x_1, x_2) = x_1^2 + x_2^2$ is differentiable at $(1, 1)$. We let $M = (\partial_{x_1} f(1, 1), \partial_{x_2} f(1, 1)) = (2, 2)$ and see that

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Mh}{\|h\|} = \lim_{h \rightarrow 0} \frac{f(1+h_1, 1+h_2) - f(1, 1) - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \quad (112)$$

$$= \lim_{h \rightarrow 0} \frac{(1+h_1)^2 + (1+h_2)^2 - 2 - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \quad (113)$$

$$= \lim_{h \rightarrow 0} \sqrt{h_1^2 + h_2^2} = 0 \quad (114)$$

So, $Df(1, 1) = (2, 2)$.

Example 1.10 (Computing Tangent Plane)

Let us find the equation of the tangent plane to $f(x, y) = \ln(2x + y)$ at $(-1, 3)$. Our total derivative, if it exists, is the covector of partials

$$Df = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right) = \left(\frac{2}{2x+y} \quad \frac{1}{2x+y} \right) \quad (115)$$

which is indeed continuous at a neighborhood of $(-1, 3)$ (in fact, in every neighborhood not containing 0). By continuity of partials, f is differentiable at $(-1, 3)$, with $Df_{(-1,3)} = (2, 1)$. The equation of the plane is then

$$z = f(-1, 3) + Df_{(-1,3)} \begin{pmatrix} x+1 \\ y-3 \end{pmatrix} = 2(x+1) + 1(y-3) \implies z = 2x + y - 1 \quad (116)$$

However, the converse is not true for either statements. You cannot just evaluate all the partial derivatives and assume that the total derivative exists!

Example 1.11 (Partial Derivatives Exist, but Not Differentiable)

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad (117)$$

At the origin $(0, 0)$, the partial derivatives are:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0. \quad (118)$$

Similarly, $\frac{\partial f}{\partial y}(0, 0) = 0$. Both partial derivatives exist. However, the total derivative **cannot** exist because the function is not continuous at $(0, 0)$. As we saw in the limits section, approaching along $y = x$ gives a limit of $1/2$, while the function value is 0. A fundamental property of differentiability is that it implies continuity; therefore, f is not differentiable at the origin.

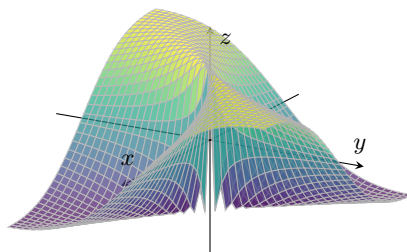


Figure 5: A function where partial derivatives exist at the origin, but the surface is “torn” there.

Example 1.12 (All Directional Derivatives Exist, but Not Differentiable)

Consider $f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. For any direction $u = (u_1, u_2)$, the directional derivative

at the origin exists and is given by $D_u f(0, 0) = \frac{u_1^2}{u_2}$ (if $u_2 \neq 0$) or 0 (if $u_2 = 0$).

Despite having directional derivatives in every single direction, the total derivative does not exist. First, f is not continuous at the origin (approaching along $y = x^2$ gives $1/2 \neq 0$). Second, even if it were continuous, the total derivative requires that the linear map $Df(a)$ approximates the function well in *every* manner of approach, not just along straight lines. Here, the “peak” of the function follows the parabola $y = x^2$, which escapes the linear approximation provided by the directional derivatives.

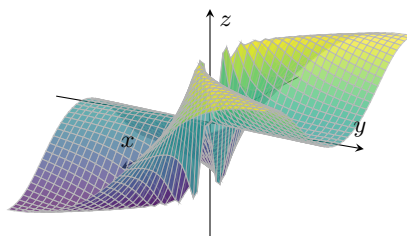


Figure 6: A function where all directional derivatives exist at the origin, yet the surface is non-differentiable.

1.3 Rules of Differentiation

Just like single variable calculus, the total derivative behaves in predictable ways: it is linear, product/quotient rules, and the chain rule.

Lemma 1.4 (Linearity of Total Derivatives)

Let $f, g : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in E$. Then, the total derivative at a is linear w.r.t. the input functions.

1. *Addition.* $D(f + g)_a = Df_a + Dg_a$
2. *Scalar Multiplication.* $D(cf)_a = cDf_a$

Proof. Let f, g be differentiable at a and $c \in \mathbb{R}$.

1. *Addition.* There exists linear functions Df_a, Dg_a and infinitesimal functions Φ_f, Φ_g such that

$$f(a + h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \quad (119)$$

$$g(a + h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \quad (120)$$

By adding the two, we get

$$(f + g)(a + h) = (f + g)(a) + (Df_a + Dg_a)(h) + (\Phi_f + \Phi_g)(h)\|h\| \quad (121)$$

where $Df_a + Dg_a$ is also linear and $\Phi_f + \Phi_g$ is also infinitesimal. So $f + g$ is differentiable with derivative $D(f + g)_a = Df_a + Dg_a$.

2. *Scalar Multiplication.* There exists a linear function Df_a and infinitesimal Φ such that

$$f(a + h) = f(a) + Df_a(h) + \Phi(h)\|h\| \quad (122)$$

By multiplying both sides by c , we get

$$(cf)(a + h) = (cf)(a) + (cDf_a)(h) + (c\Phi)(h)\|h\| \quad (123)$$

where cDf_a is also linear and Φ also infinitesimal. So, cf is differentiable with derivative $D(cf)_a = cDf_a$.

Note that for the product and quotient rules, our scope is only for scalar valued functions since products and quotients aren't defined over codomains that are vector spaces.

Lemma 1.5 (Product Rule)

Let $f, g : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a . Then,

$$D(fg)_a = Df_a g(a) + f(a) Dg_a \quad (124)$$

Proof. By differentiability, we have

$$f(a + h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \quad (125)$$

$$g(a + h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \quad (126)$$

By multiplying and expanding,^a we get

$$(fg)(a + h) = f(a)g(a) + f(a)Dg_a(h) + g(a)Df_a(h) \quad (127)$$

$$+ Df_a(h)Dg_a(h) + f(a)\Phi_g(h)\|h\| + g(a)\Phi_f(h)\|h\| + Df_a(h)\Phi_g(h)\|h\| \quad (128)$$

$$+ Dg_a(h)\Phi_f(h)\|h\| + \Phi_f(h)\|h\|\Phi_g(h)\|h\| \quad (129)$$

The map $fDg_a + Df_a f$ is linear, so it suffices to prove that the rest of the terms can be turned into a form of $\Phi(h)\|h\|$ where Φ is infinitesimal. We can see that by linearity,

$$Df_a(h)Dg_a(h) = \frac{Df_a(h)Dg_a(h)\|h\|}{\|h\|} = Df_a(h)Dg_a\left(\frac{h}{\|h\|}\right)\|h\| \quad (130)$$

where $\lim_{h \rightarrow 0} Df_a(h)Dg_a\left(\frac{h}{\|h\|}\right) = \left(\lim_{h \rightarrow 0} Df_a(h)\right)\left(\lim_{h \rightarrow 0} Dg_a\left(\frac{h}{\|h\|}\right)\right) = 0 \cdot \left(\lim_{h \rightarrow 0} Dg_a\left(\frac{h}{\|h\|}\right)\right) = 0$. By the limit rules, the rest of the terms are obvious.

^aI wish there was a cleaner solution such as the proof for the univariate case, but I wasn't able to derive one.

Lemma 1.6 (Quotient Rule)

Let $f, g : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a with $g(a) \neq 0$. Then,

$$D\left(\frac{f}{g}\right)_a = \frac{Df_a g(a) - f(a) Dg_a}{g(a)^2} \quad (131)$$

Proof. Just like the univariate case, we may make an educated guess that $D(1/g)_a = -\frac{Dg_a}{g^2}$. Therefore,

$$\frac{(1/g)(a+h) - (1/g)(a) - \left(-\frac{Dg_a}{g^2}\right)}{\|h\|} = \frac{\frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{g(a+h)g(a)^2}}{\|h\|} \quad (132)$$

$$= \frac{1}{g(a)^2} \frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} \quad (133)$$

We would like to turn this into a form that represents the limit definition of the derivative of g .

$$= \frac{1}{g(a)^2} \frac{g(a)^2 - g(a+h)^2 + g(a+h)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} \quad (134)$$

$$= \frac{1}{g(a)^2} \left(\underbrace{\frac{g(a)^2 - g(a+h)^2}{\|h\|g(a+h)}}_{\rightarrow 0} + \frac{g(a+h)[g(a+h) - g(a) - Dg_a(h)]}{\|h\|g(a+h)} \right) \quad (135)$$

The left term goes to 0 since g is differentiable and hence continuous. The right term can be simplified by canceling out the $g(a+h)$, which can be done since g is not 0 in a neighborhood of a . So, adding the limit back in, the whole term simplifies to

$$\lim_{h \rightarrow 0} \frac{(1/g)(a+h) - (1/g)(a) - \left(-\frac{Dg_a}{g^2}\right)}{\|h\|} = \lim_{h \rightarrow 0} \frac{1}{g(a)^2} \frac{g(a+h) - g(a) - Dg_a(h)}{\|h\|} \quad (136)$$

$$= \frac{1}{g(a)^2} \lim_{h \rightarrow 0} \frac{\|g(a+h) - g(a) - Dg_a(h)\|}{\|h\|} = 0 \quad (137)$$

by the definition of g being differentiable at a .

Theorem 1.7 (Chain Rule)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in E$ and $g : E' \subset \mathbb{R}^m \rightarrow \mathbb{R}^p$ be differentiable at $g(a) \in E'$. Then, $g \circ f$ is differentiable at a , and

$$D(g \circ f)_a = Dg_{f(a)} \circ Df_a \quad (138)$$

Proof. Since f is differentiable at a and g at $f(a)$, we can write

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \quad (139)$$

$$g(f(a)+t) = g(f(a)) + Dg_{f(a)}(t) + \Phi_g(t)\|t\| \quad (140)$$

Therefore, we wish to show that

$$\lim_{h \rightarrow 0} \frac{g(f(a+h)) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} = 0 \quad (141)$$

Let's remove the limit and focus on the term. First substitute $f(a+h)$ and expand the term for

$$g(f(a) + t).$$

$$= \frac{g(f(a) + \overbrace{Df_a(h) + \Phi_f(h)\|h\|}^t) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \quad (142)$$

$$= \frac{g(f(a)) + Dg_{f(a)}(Df_a(h) + \Phi_f(h)\|h\|) + \Phi_g(Df_a(h) + \Phi_f(h)\|h\|)\|h\| - g(f(a)) - Dg_{f(a)}(Df_a(h))}{\|h\|} \quad (143)$$

$$= \frac{(Dg_{f(a)} \circ Df_a)(h) + Dg_{f(a)}(\Phi_f(h)\|h\|) + \Phi_g(Df_a(h) + \Phi_f(h)\|h\|)\|h\| - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \quad (144)$$

$$= Dg_{f(a)}(\Phi_f(h)) + \Phi_g(Df_a(h) + \Phi_f(h)\|h\|) \rightarrow 0 \quad (145)$$

where the final step follows from linearity of $Dg_{f(a)}$. As $h \rightarrow 0$, the limit of the left term is 0 since linear maps are continuous, and the right term goes to 0 since Φ_g is infinitesimal. We are done.

Therefore, given the composition of function $f \circ g$, we have two methods of finding the derivative matrix of $f \circ g$ at point x_0 . First is to explicitly compute $f \circ g$ and find its $m \times p$ derivative matrix $D(f \circ g)$, and plug in a to get $D(f \circ g)_a$. The second way is to use the chain rule to find the individual total derivatives $Df_{g(a)}$ and Dg_a , and multiply them together.

1.4 Continuously Differentiable Functions

An even stronger condition beyond differentiability is continuous partials, and we often prove continuity of partials to prove differentiability.

Definition 1.7 (Continuously Differentiable)

A function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **continuously differentiable** at $a \in E$ if it satisfies either of the equivalent definitions:

1. f is differentiable at a and the map $a \in \mathbb{R}^n \rightarrow Df_a \in \mathbb{R}^{m \times n}$ is continuous.
2. f is differentiable at a and all the partial derivatives $a \in \mathbb{R}^n \rightarrow \frac{\partial f_j}{\partial x_i}(a) \in \mathbb{R}$ is continuous.

Proof. We prove bidirectionally.

1. (1 $\rightarrow$ 2). Let $\phi_{ij} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ be the projection of the matrix onto the ij th element. This is continuous and we can see that

$$\frac{\partial f_j}{\partial x_i}(a) = \phi_{ij}(Df_a) \quad (146)$$

so all partial derivatives are continuous.

2. (2 $\rightarrow$ 1). Let $\epsilon > 0$. If all partial derivatives are continuous, then there exists a $\delta > 0$ such that

$$\|x - a\| < \delta \implies \left\| \frac{\partial f_j}{\partial x_i}(x) - \frac{\partial f_j}{\partial x_i}(a) \right\| < \frac{\epsilon}{mn} \implies \|Df_a\|_1 = \sum_{i,j} (Df_a)_{ij} < \epsilon \quad (147)$$

So $a \rightarrow Df_a$ is continuous with respect to the L_1 norm on $\mathbb{R}^{m \times n}$. But since all L_p norms are equivalent in finite-dimensional Euclidean space, it is continuous with respect to all norms.

Theorem 1.8 (Continuous Partial $\implies$ Differentiability)

Given a function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ and a point $a \in E$, if all the partials $\partial_{x_i} f$ exist and are continuous at a , then f is differentiable at a .

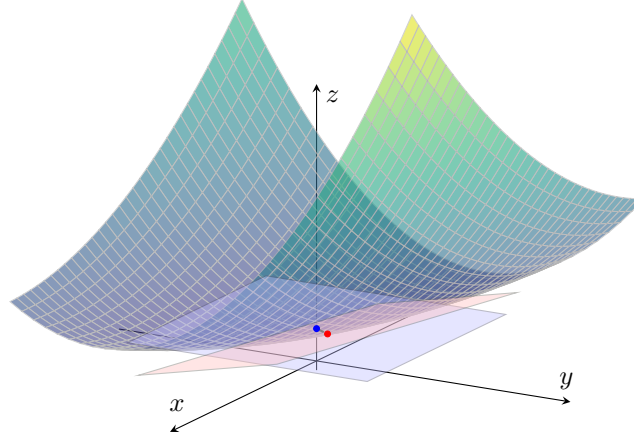


Figure 7: We can also visualize this theorem. Since the partials are continuous, then the tangent subspace, which is determined by the span of the tangent vectors determined by the partials, also changes continuously, and therefore the total derivative within a neighborhood of a exists.

Proof. The idea is to bound the perturbations in each component using the mean value theorem. Let us consider real valued $f : \mathbb{R}^n \rightarrow \mathbb{R}$ for now. Given $a, a + h \in \mathbb{R}^n$, let us construct a stepwise path from a to $a + h$. Let us denote $\phi_k(h) = (h_1, h_2, \dots, h_k, 0, \dots, 0)$. Then, we can construct the sequence

$$a = a + \phi_0(h), a + \phi_1(h), a + \phi_2(h), \dots, a + \phi_k(h) = a + h \quad (148)$$

We take the matrix of partials

$$A = \left(\frac{\partial f}{\partial x_1}(a) \quad \dots \quad \frac{\partial f}{\partial x_n}(a) \right) \quad (149)$$

and wish to show that this is indeed the total derivative. We see that

$$f(a + h) - f(a) - Ah = f(a + h) - f(a) - \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) h_i \quad (150)$$

$$= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) h_i + (f(a + \phi_i(h)) - f(a + \phi_{i-1}(h))) \quad (151)$$

Since $h_i \mapsto f(a + \phi_i(h)) - f(a + \phi_{i-1}(h))$ is a univariate differentiable function, by MVT, there exists a $c_i \in (a_i, a_i + h_i)$ s.t. $\frac{\partial f}{\partial x_i}(c_i) = \frac{f(a + \phi_i(h)) - f(a + \phi_{i-1}(h))}{h_i}$. Therefore, the above equals

$$= \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) h_i \quad (152)$$

So,

$$\|f(a + h) - f(a) - Ah\| = \left\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) h_i \right\| \quad (153)$$

Since $|h_i| \leq \|h\|$,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \left\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) \frac{h_i}{\|h\|} \right\| \quad (154)$$

$$= \left\| \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right) \right\| \quad (155)$$

$$\leq \sum_{i=1}^n \left\| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right\| \quad (156)$$

But from continuity of partials, we know that for $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$\|h\| < \delta \implies \left| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(a + \phi_k(h)) \right| < \frac{\epsilon}{n} \quad (157)$$

$$\implies \left| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right| < \frac{\epsilon}{n} \quad (158)$$

since $c_i \in (a, a + \phi_k(h))$. Therefore,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \sum_{i=1}^n \left| \frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i) \right| < \epsilon \quad (159)$$

While continuous partial derivatives guarantee differentiability, the converse is not true. A function can be differentiable even if its partial derivatives are not continuous. However, if they are not continuous, we cannot use the previous theorem to determine differentiability and must revert to the definition.

Example 1.13 (Non-Continuous Partials and Not Differentiable)

Recall $f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. The partial derivative $\frac{\partial f}{\partial x}$ for $(x, y) \neq (0, 0)$ is:

$$\frac{\partial f}{\partial x} = \frac{y(x^2 + y^2) - xy(2x)}{(x^2 + y^2)^2} = \frac{y^3 - x^2y}{(x^2 + y^2)^2}.$$

At the origin, we found $\frac{\partial f}{\partial x}(0, 0) = 0$. However, $\lim_{(x,y) \rightarrow (0,0)} \frac{y^3 - x^2y}{(x^2 + y^2)^2}$ does not exist (it depends on the path). Thus, the partial derivatives are not continuous at the origin. As shown before, the function is not differentiable because it is not continuous.

Example 1.14 (Non-Continuous Partials and Differentiable)

Consider $f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. At the origin, the partial derivatives exist and are 0:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/|h|) - 0}{h} = \lim_{h \rightarrow 0} h \sin(1/|h|) = 0.$$

Away from the origin, $\frac{\partial f}{\partial x}$ involves a $\cos(1/\sqrt{x^2 + y^2}) \cdot \frac{1}{x^2 + y^2}$ term which oscillates wildly as $(x, y) \rightarrow (0, 0)$. Thus, the partials are not continuous.

However, f is differentiable at the origin. Using the definition with $Df(0,0) = 0$:

$$\lim_{h \rightarrow 0} \frac{|(h_1^2 + h_2^2) \sin(1/\|h\|) - 0 - 0|}{\|h\|} = \lim_{h \rightarrow 0} \|h\| \sin(1/\|h\|) = 0.$$

This satisfies the definition of the total derivative.

Theorem 1.9 (C^1 Space)

The set of all continuously differentiable functions $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$, denoted $C^1(E)$, is a vector space.

Proof. The proof is trivial since Df is continuous. Therefore, sums and scalar multiples of are continuous as well.

Note that from now, whenever we talk about differentiating a function f , we will assume that it is C^1 . This is overkill, since the set of all k -times differentiable functions is a subset of C^k , but it is conventional to work with C^k functions.

1.5 Mean Value Theorem

Lemma 1.10 (Mean Value Theorem for Paths)

If $f : E \subset \mathbb{R} \rightarrow \mathbb{R}^m$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $x \in (a, b)$ s.t.

$$\|f(b) - f(a)\| \leq (b - a)\|f'(x)\| \quad (160)$$

Proof. The general idea is that we want to convert this into a scalar-valued function in a clever way.

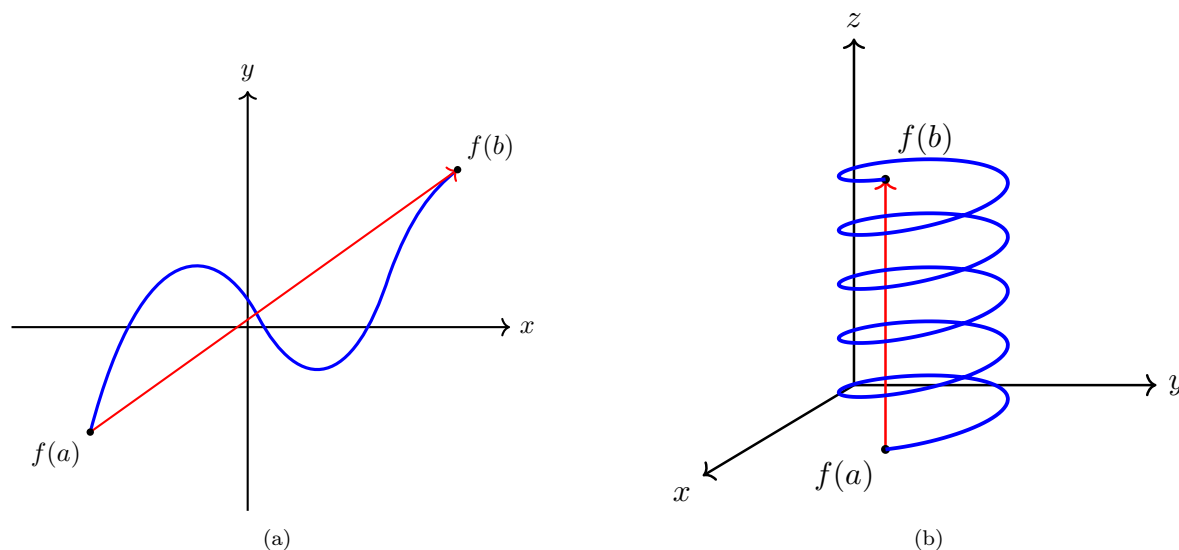


Figure 8: If we sketch out a path in $\mathbb{R}^2$, we might speculate that at some point, the tangent line of the curve must be parallel to the secant vector $f(b) - f(a)$. Perhaps defining $\phi(x) = \langle f(b) - f(a), \frac{Df_x}{\|Df_x\|} \rangle$ and trying to maximize it using EVT or something might help. However, there is no guarantee that Df_x is continuous, and unfortunately, this assumption turns out not to be true as seen in the solenoid.

After some thought (see figure caption), we might think of looking at the inner product of the actual

vectors on the curve with the secant vector, defining

$$\phi(x) = \langle f(b) - f(a), f(x) \rangle \quad (161)$$

ϕ is differentiable on (a, b) and continuous on $[a, b]$, so by MVT, there exists some $x \in (a, b)$ such that

$$\phi(b) - \phi(a) = D\phi_x(b - a) = \langle f(b) - f(a), Df_x \rangle (b - a) \quad (162)$$

But $\phi(b) - \phi(a) = \langle f(b) - f(a), f(b) - f(a) \rangle = \|f(b) - f(a)\|^2$, so by Cauchy Schwarz, we get

$$\|f(b) - f(a)\|^2 = \langle f(b) - f(a), Df_x \rangle (b - a) \leq \|f(b) - f(a)\| \|Df_x\| (b - a) \quad (163)$$

Therefore, $\|f(b) - f(a)\| \leq \|Df_x\| (b - a)$.

There is a nice reason why equality doesn't hold in higher dimensions. For my first attempt at this proof, I tried to replicate my work for proving that C^1 implies differentiable by dividing f into component-wise paths. I defined

$$\phi_k(f) = (f_1(a), \dots, f_k(a), f_{k+1}(b), \dots, f_m(b))^T \quad (164)$$

Therefore, I was hoping to bound $\|f(b) - f(a)\| \leq \sum_{k=1}^m \|\phi_k(f) - \phi_{k-1}(f)\| = \sum_{k=1}^m |f_k(b) - f_k(a)|$. But invoking the univariate MVT for the real-valued function f_k leads to finding multiple $x_k \in (a, b)$. This is a problem since we need to find one x .

Example 1.15 (MVT Inequality Does Not Hold on Circle)

Let

$$f : [0, 2\pi] \rightarrow \mathbb{R}^2, \quad f(x) = (\cos(x), \sin(x)) \quad (165)$$

Then, $f(2\pi) - f(0) = 0$, but

$$Df_x = \begin{pmatrix} -\sin(x) \\ \cos(x) \end{pmatrix} \quad (166)$$

is never 0 over $x \in [0, 2\pi]$.

Theorem 1.11 (Mean Value Theorem)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable on convex open set E . If there exists a $M \in \mathbb{R}$ s.t. $\|Df_x\| \leq M$ for all $x \in E$, then

$$\|f(b) - f(a)\| \leq M \|b - a\| \quad (167)$$

for all $a, b \in E$.

Proof. Since E is convex, for any $a, b \in E$, we can consider

$$\gamma : [0, 1] \rightarrow E, \quad \gamma(t) := ta + (1 - t)b \quad (168)$$

Then, $f \circ \gamma$ is a path function that is continuous on $[0, 1]$ and differentiable on $(0, 1)$. Therefore, there exists a $t_0 \in (0, 1)$ (and hence, corresponding $x_0 = \gamma(t_0) \in E$) such that

$$\|(f \circ \gamma)(1) - (f \circ \gamma)(0)\| \leq \|D(f \circ \gamma)_{t_0}\| \|1 - 0\| \quad (169)$$

$$= \|Df_{x_0} D\gamma_{t_0}\| \quad (170)$$

$$\leq \|Df_{x_0}\| \|a - b\| \quad (171)$$

$$\leq \|Df_{x_0}\| \|a - b\| \quad (172)$$

$$\leq M \|b - a\| \quad (173)$$

where in the second step, we used the chain rule, in the third step, we computed the derivative $D\gamma = a - b$, and in the fourth step, we used the operator norm bound. The left hand side is really $\|f(b) - f(a)\|$, and we have our desired result.

The following corollary is pretty obvious.

Corollary 1.12 (Vanishing Derivative Implies Constant)

Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable on convex open set E . If $\|Df_x\| = 0$ for all $x \in E$, then f is constant on E .

Proof. Set $M = 0$ in the proof above.

1.6 Inverse Function Theorem

Definition 1.8 (Contraction)

Given a metric space (X, d) and some $c < 1$, a **contraction** is a mapping $\phi : X \rightarrow X$ such that

$$d(\phi(x), \phi(y)) \leq cd(x, y) \quad (174)$$

Theorem 1.13 (Banach Fixed Point Theorem)

Given a complete metric space (X, d) and a contraction ϕ on X , there exists a unique point $x \in X$ such that $\phi(x) = x$.

Proof. We give a constructive proof on how to find such a point. Let us pick any $x_0 \in X$ and set $x_k = \phi^k(x_0)$ to be the output of the k -fold composition of ϕ . Then,

$$d(x_k, x_{k+1}) \leq c^k d(x_0, x_1) \quad (175)$$

So given any $m, n \in \mathbb{N}$, WLOG let $m < n$. Then,

$$d(x_m, x_n) \leq \sum_{k=m}^{n-1} d(x_k, x_{k+1}) \quad (176)$$

$$\leq \sum_{k=m}^{n-1} c^k d(x_0, x_1) \quad (177)$$

$$= d(x_0, x_1) \sum_{k=m}^{n-1} c^k \quad (178)$$

$$= d(x_0, x_1) \sum_{k=m}^{\infty} c^k \quad (179)$$

Since the series converges from the geometric test, the tail sum $\sum_{k=m}^{\infty} c^k$ converges to 0 as $m \rightarrow +\infty$. So, $(x_n)_n$ is a Cauchy sequence and therefore converges to some $x \in X$.

Since ϕ is a contraction, ϕ is continuous. In fact it is uniformly continuous, since given $\epsilon > 0$, we can set $\delta = \frac{\epsilon}{c}$ and see that for all $x, y \in X$,

$$d(x, y) < \delta = \frac{\epsilon}{c} \implies d(\phi(x), \phi(y)) < c \frac{\epsilon}{c} = \epsilon \quad (180)$$

Therefore,

$$\phi(x) = \lim_{n \rightarrow +\infty} \phi(x_n) = \lim_{n \rightarrow +\infty} x_{n+1} = x \quad (181)$$

To prove uniqueness, assume that $x, y \in X$ are both fixed points. Then,

$$d(x, y) = d(\phi(x), \phi(y)) \leq cd(x, y) \implies d(x, y) = 0 \implies x = y \quad (182)$$

It is a bit strange that we just introduce this fixed point theorem all of a sudden when talking about invertibility, but this becomes useful in the proof for the inverse function theorem. When we have a point y and want to prove an inverse x exists such that $y = f(x)$, we can use the fixed point theorem to guarantee existence of such a point x .

Theorem 1.14 (Inverse Function Theorem)

Let $U \subset \mathbb{R}^n$ be an open neighborhood of a , and let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 function. If Df_a is invertible, then

1. there exists open neighborhoods V of a and W of $f(a)$, with $V \subset U$, such that

$$f|_V : V \rightarrow W \quad (183)$$

is a bijection.

2. Moreover, the inverse map $(f|_V)^{-1} : W \rightarrow V$ is C^1 , and for every $y \in W$,

$$D(f^{-1})_y = (Df_{f^{-1}(y)})^{-1}. \quad (184)$$

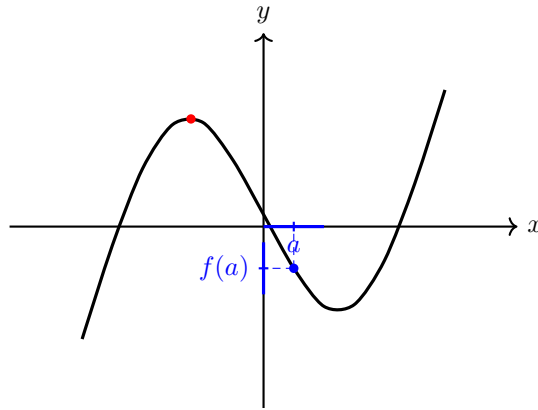


Figure 9: A cubic can fail to be locally invertible at a turning point, but behave regularly at nearby non-turning points.

Proof. The idea is to linearize $f(x)$. By definition of differentiability at $x \in E$, there exists a linear map $Df_a : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and infinitesimal function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $f(x) = f(a) + Df_a(x - a) + \Phi(x - a)\|x - a\|$ within some neighborhood of x . Therefore, we can approximate

$$f(x) \approx f(a) + Df_a(x - a) \quad (185)$$

Given our center $(a, f(a))$ and $x \in \mathbb{R}^n$, we can use this formula to approximate $f(x)$. We must go backwards and use $y = f(x)$ to approximate x . We can write the inverse of this affine function as $x = a + Df_a^{-1}(f(x) - f(a))$. So for all $y_0 \in \mathbb{R}^n$, we define the function which attempts to estimate the inverse of y_0 with respect to center $(x, f(x))$.

$$\phi_{y_0} : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad \phi_{y_0}(x) := x + Df_a^{-1}(y_0 - f(x)) \quad (186)$$

The idea is that given some $y_0 \in \mathbb{R}^n$, we are trying to estimate the preimage of y_0 , so if we set the center itself as $f^{-1}(y_0)$, then we should get back exactly the preimage. This is indeed the case:

$$\phi_{y_0}(f^{-1}(y_0)) = f^{-1}(y_0) + Df_a^{-1}(y_0 - f(f^{-1}(y_0))) = f^{-1}(y_0) \quad (187)$$

The idea is to prove that ϕ_{y_0} is a contraction of some subset that is a complete metric space, and so (by the Banach fixed point theorem) the unique fixed point—call it $x_0 \in \mathbb{R}^n$ —is the inverse. Let's do this step by step.

1. ϕ_0 is a contraction. $\phi_{y_0}(x) = x + Df_a^{-1}(y_0 - f(x))$, which implies that

$$D(\phi_{y_0})_x = I - Df_a^{-1}Df_x = Df_a^{-1}(Df_a - Df_x) \quad (188)$$

But since $x \rightarrow Df_x$ is continuous at a , there exists an open neighborhood U containing a s.t.

$$x \in U \implies \|Df_x - Df_a\| < \frac{1}{2\|Df_a^{-1}\|} \quad (189)$$

Therefore, when $x \in U$,

$$\|D(\phi_{y_0})_x\| = \|Df_a^{-1}\|\|Df_a - Df_x\| \quad (190)$$

$$\leq \|Df_a^{-1}\|\frac{1}{2\|Df_a^{-1}\|} = \frac{1}{2} \quad (191)$$

Therefore, by the MVT, for all $x_1, x_2 \in U$,

$$\|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \quad (192)$$

and so, ϕ_{y_0} is a contraction on U .

2. f is injective over U . Say that x_1, x_2 are both fixed points of U . Then,

$$\|x_1 - x_2\| = \|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \implies x_1 = x_2 \quad (193)$$

But note that

$$f(x_0) = y_0 \iff \phi_{y_0}(x_0) = x_0 + Df_a^{-1}(y_0 - f(x_0)) = x_0 \quad (194)$$

$$\iff x_0 \text{ is a fixed point of } \phi_{y_0} \quad (195)$$

So, since there is at most 1 fixed point, there can be at most one $x_0 \in U$ s.t. $f(x_0) = y_0$. Thus, f is injective and therefore $f : U \rightarrow f(U)$ is bijective, hence is invertible. Note that we did *not* use Banach fixed point theorem yet.

3. $V = f(U)$ is open. This is much trickier than it looks. First, pick $y_1 \in V$ and so $y_1 = f(x_1)$ for some $x_1 \in U$. We wish to show that there exists a δ -neighborhood around y_1 , which establishes that V is open. The choose δ , let us choose $\epsilon > 0$ so small that $\overline{B_\epsilon(x_1)} \subset U$. Then, if $x \in B_\epsilon(x_1)$,

$$\|\phi_{y_0}(x) - x_1\| \leq \|\phi_{y_0}(x) - \phi_{y_0}(x_1)\| + \|\phi_{y_0}(x_1) - x_1\| \quad (196)$$

$$\leq \frac{1}{2}\|x - x_0\| + \|Df_a^{-1}(y_0 - f(x_1))\| \quad (197)$$

$$\leq \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - f(x_1)\| \quad (198)$$

$$= \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - y_1\| \quad (199)$$

Therefore, this hints that we need to bound $\|y_0 - y_1\|$, so we can choose y_0 (and hence ϕ_{y_0}) s.t.

$$\|y_0 - y_1\| < \frac{\epsilon}{2\|Df_a^{-1}\|} \quad (200)$$

This implies that $\|\phi_{y_0}(x) - x_1\| < \epsilon$. That is, as long as the center x is close enough to x_1 and y_0 is close enough to y_1 , the approximate inverse of y_0 is close to x_1 . Therefore, $\phi_{y_0}(x) \in B_\epsilon(x_1)$. So, if $x \in B_\epsilon(x_1)$ and $\|y_0 - y_1\| < \frac{\epsilon}{2\|Df_a^{-1}\|}$, then $\phi_{y_0}(x) \in B_\epsilon(x_1)$. Since ϕ_{y_0} is a contraction over U , it is also a contraction over $\overline{B_\epsilon(x_1)}$.

Now since $\overline{B_\epsilon(x_1)}$ is closed in $\mathbb{R}^n$, it is complete, so by Banach fixed point theorem, ϕ_{y_0} has a unique fixed point, which turns out to be $x_0 \in \overline{B_\epsilon(x_1)}$. So, $y_0 = f(x_0)$, and therefore

$$x_0 \in \overline{B_\epsilon(x_1)} \implies y_0 = f(x_0) \in f(B_\epsilon(x_1)) \subset f(U) = V \quad (201)$$

And so, by choosing $y_0 \in B_{\epsilon/(2\|Df_a^{-1}\|)}(y_1)$, we have $y_0 \in V \implies B_{\epsilon/(2\|Df_a^{-1}\|)}(y_1) \subset V$.

Great, so we have established that $f : U \rightarrow V$ is bijective and thus invertible. Now we wish to show that for $f : U \rightarrow V$, $D(f^{-1})_b = (Df_a)^{-1}$.

Example 1.16 (A Univariate Example)

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = x^2. \quad (202)$$

Its derivative is

$$Df_x = f'(x) = 2x. \quad (203)$$

Therefore, at any point $x_0 \neq 0$, we have

$$Df_{x_0} = 2x_0 \neq 0. \quad (204)$$

Hence Df_{x_0} is invertible as a linear map from $\mathbb{R}$ to $\mathbb{R}$.

By the inverse function theorem, for every $x_0 \neq 0$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \rightarrow f(V) \quad (205)$$

is invertible.

For example, at $x_0 = 1$, we can take a small interval around 1, such as

$$V = \left(\frac{1}{2}, \frac{3}{2}\right). \quad (206)$$

On this interval, $f(x) = x^2$ is injective, so it has a well-defined inverse

$$f^{-1}(y) = \sqrt{y}. \quad (207)$$

However, f is not invertible on all of $\mathbb{R}$, since

$$f(1) = 1 = f(-1). \quad (208)$$

Thus the inverse function theorem guarantees local invertibility near points where $f'(x_0) \neq 0$, but it does not guarantee global invertibility.

Example 1.17

Consider the vector-valued function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$f(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \end{pmatrix}$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) \end{pmatrix} \implies \det Jf(x_1, x_2) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2) = e^{2x_1}$$

Since the determinant e^{2x_1} is nonzero everywhere, Df_x is nonsingular. Thus, the theorem guarantees that for every point $x_0 \in \mathbb{R}^2$, there exists a neighborhood about x_0 over which f is invertible. However, this does not mean f is invertible over its entire domain: in this case f isn't even injective since it is periodic: e.g. the preimage of $(e, 0)$ contains $(1, 0)$ and $(1, 2\pi)$.

Example 1.18 (A Function from $\mathbb{R}^3$ to $\mathbb{R}^3$)

Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$f(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \\ x_3 \end{pmatrix}. \quad (209)$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) & 0 \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (210)$$

Therefore,

$$\det Jf(x_1, x_2, x_3) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2). \quad (211)$$

Hence

$$\det Jf(x_1, x_2, x_3) = e^{2x_1}. \quad (212)$$

Since $e^{2x_1} \neq 0$ for all $x_1 \in \mathbb{R}$, the derivative Df_x is invertible at every point $x \in \mathbb{R}^3$.

Thus, by the inverse function theorem, for every point $x_0 \in \mathbb{R}^3$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \rightarrow f(V) \quad (213)$$

is invertible.

However, f is not invertible on all of $\mathbb{R}^3$, since it is periodic in the x_2 -variable. For example,

$$f(1, 0, 5) = \begin{pmatrix} e \\ 0 \\ 5 \end{pmatrix} = f(1, 2\pi, 5). \quad (214)$$

Therefore, f is locally invertible everywhere, but not globally invertible.

1.7 Implicit Function Theorem

Remember that given an explicit representation of a set $y = f(x)$, we can easily find the implicit form as $F(x, y) = y - f(x) = 0$. What about the other way around? That is, given an implicit representation of some surface, what conditions must be met so that it can be represented as the graph of a function? The implicit function theorem is a tool that allows relations between points in $\mathbb{R}^n$ to be converted to functions of several real variables. That is, it states that for sufficiently "nice" points on a n -dimensional surface defined as $F(x, y) = 0$ (where $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$), we can locally pretend that this surface is a graph of a function $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ whose graph $(x, g(x))$ is precisely the set of all (x, y) s.t. $f(x, y) = 0$. When $m = 1$, it basically states that if an implicit surface suffices the vertical line test in a neighborhood, then it can be written as a function.

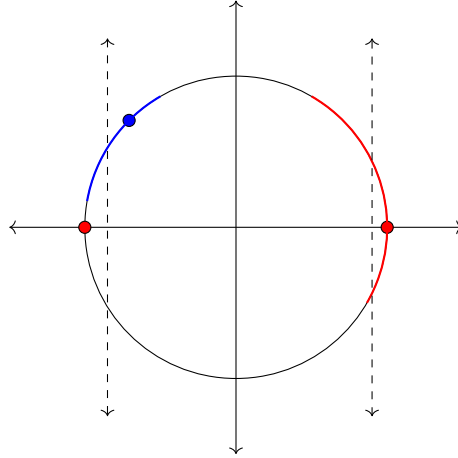


Figure 10: Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x^2 + y^2 - 1$. The level set at $z = 0$ would be the set of points satisfying $x^2 + y^2 - 1 = 0$, i.e. the unit circle. The derivative of f with respect to y is 0 at the points $(-1, 0)$ and $(1, 0)$, meaning that in any neighborhood of these points, we cannot define a function of y with respect to x . This is true, indeed, since any such function would fail the vertical line test, which can be seen in the red neighborhood around $(1, 0)$. However, the blue neighborhood of the point $(-\sqrt{2}/2, \sqrt{2}/2)$ does indeed define a function of y with respect to x satisfying the vertical line test.

Theorem 1.15 (Implicit Function Theorem)

Let $F : \mathbb{R}^n \times \mathbb{R}^c \rightarrow \mathbb{R}^c$ be a C^1 function. Write points in $\mathbb{R}^n \times \mathbb{R}^c$ as (x, y) , where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^c$. If

1. $F(a, b) = 0$, and
2. $D_y F_{(a, b)}$ is invertible,

then there exist open neighborhoods U of a and V of b , and a unique C^1 function

$$h : U \rightarrow V \quad (215)$$

such that $h(a) = b$ and, for every $(x, y) \in U \times V$,

$$F(x, y) = 0 \iff y = h(x). \quad (216)$$

Moreover, for every $x \in U$,

$$Dh_x = -(D_y F_{(x, h(x))})^{-1} D_x F_{(x, h(x))}. \quad (217)$$

Proof. Define a function

$$G : \mathbb{R}^n \times \mathbb{R}^c \rightarrow \mathbb{R}^n \times \mathbb{R}^c \quad (218)$$

by

$$G(x, y) = (x, F(x, y)). \quad (219)$$

Since $F(a, b) = 0$, we have

$$G(a, b) = (a, 0). \quad (220)$$

We compute the derivative of G . Since the first component of G is just x , and the second component is $F(x, y)$, we have

$$DG_{(x, y)} = \begin{pmatrix} I_n & 0 \\ D_x F_{(x, y)} & D_y F_{(x, y)} \end{pmatrix}. \quad (221)$$

In particular,

$$DG_{(a, b)} = \begin{pmatrix} I_n & 0 \\ D_x F_{(a, b)} & D_y F_{(a, b)} \end{pmatrix}. \quad (222)$$

This is a block lower triangular matrix. Therefore,

$$\det DG_{(a,b)} = \det(I_n) \det(D_y F_{(a,b)}) = \det(D_y F_{(a,b)}). \quad (223)$$

Since $D_y F_{(a,b)}$ is invertible, we have

$$\det DG_{(a,b)} \neq 0. \quad (224)$$

Hence $DG_{(a,b)}$ is invertible.

By the inverse function theorem, there exist open neighborhoods N of (a, b) and P of $(a, 0)$ such that

$$G|_N : N \longrightarrow P \quad (225)$$

is a bijection with a C^1 inverse

$$(G|_N)^{-1} : P \longrightarrow N. \quad (226)$$

Since N is an open neighborhood of (a, b) , we may choose open neighborhoods U_0 of a and V of b such that

$$U_0 \times V \subset N. \quad (227)$$

Since P is an open neighborhood of $(a, 0)$, we may choose an open neighborhood U_1 of a such that

$$U_1 \times \{0\} \subset P. \quad (228)$$

For $x \in U_1$, define

$$h(x) = \pi_y((G|_N)^{-1}(x, 0)), \quad (229)$$

where π_y denotes projection onto the $\mathbb{R}^c$ component. In other words, $h(x)$ is the unique vector such that

$$(G|_N)^{-1}(x, 0) = (x, h(x)). \quad (230)$$

Since $(G|_N)^{-1}$ is C^1 , the function h is also C^1 .

Also,

$$h(a) = b, \quad (231)$$

because

$$G(a, b) = (a, 0). \quad (232)$$

Shrinking U_1 if necessary, we may assume that

$$U_1 \subset U_0 \quad (233)$$

and

$$h(U_1) \subset V. \quad (234)$$

Set

$$U = U_1. \quad (235)$$

We now show that $F(x, y) = 0$ is equivalent to $y = h(x)$ on $U \times V$. First, if $x \in U$, then

$$G(x, h(x)) = (x, 0). \quad (236)$$

Since

$$G(x, h(x)) = (x, F(x, h(x))), \quad (237)$$

it follows that

$$F(x, h(x)) = 0. \quad (238)$$

Conversely, suppose that $(x, y) \in U \times V$ and

$$F(x, y) = 0. \quad (239)$$

Then

$$G(x, y) = (x, 0). \quad (240)$$

But also

$$G(x, h(x)) = (x, 0). \quad (241)$$

Since $(x, y) \in U \times V \subset N$ and $(x, h(x)) \in U \times V \subset N$, and since $G|_N$ is injective, we conclude that

$$(x, y) = (x, h(x)). \quad (242)$$

Therefore,

$$y = h(x). \quad (243)$$

Hence, for every $(x, y) \in U \times V$,

$$F(x, y) = 0 \quad \Longleftrightarrow \quad y = h(x). \quad (244)$$

This also proves uniqueness. Indeed, if another function $\tilde{h} : U \rightarrow V$ satisfies

$$F(x, \tilde{h}(x)) = 0 \quad (245)$$

for all $x \in U$, then by the equivalence above,

$$\tilde{h}(x) = h(x) \quad (246)$$

for all $x \in U$.

It remains to compute the derivative of h . Since

$$F(x, h(x)) = 0 \quad (247)$$

for every $x \in U$, we differentiate both sides with respect to x . By the chain rule,

$$D_x F_{(x, h(x))} + D_y F_{(x, h(x))} Dh_x = 0. \quad (248)$$

Therefore,

$$D_y F_{(x, h(x))} Dh_x = -D_x F_{(x, h(x))}. \quad (249)$$

Since $D_y F_{(a, b)}$ is invertible, and invertibility is an open condition, we may shrink U and V if necessary so that $D_y F_{(x, h(x))}$ is invertible for every $x \in U$. Thus,

$$Dh_x = - \left(D_y F_{(x, h(x))} \right)^{-1} D_x F_{(x, h(x))}. \quad (250)$$

This proves the theorem.

Example 1.19 (Circle Example)

Let $n = c = 1$, and consider the function $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x_1, x_2) = x_1^2 + x_2^2 - 1. \quad (251)$$

The level set

$$F(x_1, x_2) = 0 \quad (252)$$

is the unit circle

$$x_1^2 + x_2^2 = 1. \quad (253)$$

We would like to determine at which points of the circle we can locally write x_2 as a function of x_1 .

The total derivative of F is

$$DF_{(x_1, x_2)} = (\partial_{x_1} F \quad \partial_{x_2} F) = (2x_1 \quad 2x_2). \quad (254)$$

In this case, if we want to solve for x_2 as a function of x_1 , then we look at the partial derivative with respect to x_2 :

$$D_{x_2} F_{(x_1, x_2)} = 2x_2. \quad (255)$$

This is invertible as a 1×1 matrix precisely when

$$x_2 \neq 0. \quad (256)$$

Therefore, by the implicit function theorem, at every point of the circle with $x_2 \neq 0$, we can locally write the circle in the form

$$x_2 = g(x_1). \quad (257)$$

For example, consider the point

$$\left(\frac{4}{5}, \frac{3}{5}\right) \quad (258)$$

on the unit circle. Since $\frac{3}{5} \neq 0$, we can locally solve for x_2 as a function of x_1 . Near this point, the appropriate branch is

$$x_2 = g(x_1) = \sqrt{1 - x_1^2}. \quad (259)$$

On the other hand, at the points $(1, 0)$ and $(-1, 0)$, we have

$$D_{x_2} F_{(x_1, x_2)} = 2x_2 = 0. \quad (260)$$

So the implicit function theorem does not allow us to solve for x_2 as a function of x_1 near these points. Indeed, near $x_1 = 1$ or $x_1 = -1$, the circle fails to be the graph of a function of x_1 , since nearby x_1 -values correspond to two possible x_2 -values.

The derivative of the implicit function g is given by the theorem:

$$Dg = -(D_{x_2} F)^{-1} D_{x_1} F. \quad (261)$$

Therefore,

$$Dg = -(2x_2)^{-1}(2x_1) = -\frac{x_1}{x_2}. \quad (262)$$

Since $x_2 = g(x_1)$, this becomes

$$g'(x_1) = -\frac{x_1}{g(x_1)}. \quad (263)$$

Similarly, if we want to solve for x_1 as a function of x_2 , then we look at the partial derivative with respect to x_1 :

$$D_{x_1} F_{(x_1, x_2)} = 2x_1. \quad (264)$$

This is invertible precisely when

$$x_1 \neq 0. \quad (265)$$

Hence, at every point of the circle with $x_1 \neq 0$, we can locally write the circle in the form

$$x_1 = h(x_2). \quad (266)$$

The derivative of h is

$$Dh = -(D_{x_1} F)^{-1} D_{x_2} F. \quad (267)$$

Therefore,

$$Dh = -(2x_1)^{-1}(2x_2) = -\frac{x_2}{x_1}. \quad (268)$$

Since $x_1 = h(x_2)$, this becomes

$$h'(x_2) = -\frac{x_2}{h(x_2)}. \quad (269)$$

Example 1.20 (Sphere Example)

Let $n = 2$ and $c = 1$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x_1, x_2, y) = x_1^2 + x_2^2 + y^2 - 1. \quad (270)$$

The level set

$$F(x_1, x_2, y) = 0 \quad (271)$$

is the unit sphere

$$x_1^2 + x_2^2 + y^2 = 1. \quad (272)$$

We would like to determine at which points of the sphere we can locally write y as a function of (x_1, x_2) . The total derivative of F is

$$DF_{(x_1, x_2, y)} = (\partial_{x_1} F \quad \partial_{x_2} F \quad \partial_y F) = (2x_1 \quad 2x_2 \quad 2y). \quad (273)$$

If we want to solve for y as a function of (x_1, x_2) , then we look at the partial derivative with respect to y :

$$D_y F_{(x_1, x_2, y)} = 2y. \quad (274)$$

This is invertible as a 1×1 matrix precisely when

$$y \neq 0. \quad (275)$$

Therefore, by the implicit function theorem, at every point of the sphere with $y \neq 0$, we can locally write the sphere in the form

$$y = g(x_1, x_2). \quad (276)$$

For example, consider the point

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right) \quad (277)$$

on the unit sphere. Since $\frac{1}{\sqrt{2}} \neq 0$, we can locally solve for y as a function of (x_1, x_2) . Near this point, the appropriate branch is

$$y = g(x_1, x_2) = \sqrt{1 - x_1^2 - x_2^2}. \quad (278)$$

On the other hand, at points on the equator where $y = 0$, the implicit function theorem does not allow us to solve for y as a function of (x_1, x_2) . Geometrically, near such points, the sphere has both an upper and a lower branch over the same nearby (x_1, x_2) -values.

The derivative of the implicit function g is given by

$$Dg = -(D_y F)^{-1} D_x F. \quad (279)$$

In this case,

$$D_x F = (2x_1 \quad 2x_2) \quad (280)$$

and

$$D_y F = 2y. \quad (281)$$

Therefore,

$$Dg = -(2y)^{-1} (2x_1 \quad 2x_2) = \left(-\frac{x_1}{y} \quad -\frac{x_2}{y} \right). \quad (282)$$

Since $y = g(x_1, x_2)$, this becomes

$$Dg_{(x_1, x_2)} = \left(-\frac{x_1}{g(x_1, x_2)} \quad -\frac{x_2}{g(x_1, x_2)} \right). \quad (283)$$

Example 1.21 (A System of Two Equations)

Let $n = 2$ and $c = 2$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$F(x_1, x_2, y_1, y_2) = \begin{pmatrix} y_1^2 + y_2 - x_1 \\ y_1 + e^{y_2} - x_2 \end{pmatrix}. \quad (284)$$

We would like to locally solve the system

$$F(x_1, x_2, y_1, y_2) = 0 \quad (285)$$

for (y_1, y_2) as a function of (x_1, x_2) .

First observe that

$$F(1, 2, 1, 0) = \begin{pmatrix} 1^2 + 0 - 1 \\ 1 + e^0 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (286)$$

Thus the point

$$(x_1, x_2, y_1, y_2) = (1, 2, 1, 0) \quad (287)$$

lies on the zero set of F .

The derivative of F with respect to the y -variables is

$$D_y F_{(x_1, x_2, y_1, y_2)} = \begin{pmatrix} \partial_{y_1} F_1 & \partial_{y_2} F_1 \\ \partial_{y_1} F_2 & \partial_{y_2} F_2 \end{pmatrix} = \begin{pmatrix} 2y_1 & 1 \\ 1 & e^{y_2} \end{pmatrix}. \quad (288)$$

At the point $(1, 2, 1, 0)$, this becomes

$$D_y F_{(1, 2, 1, 0)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \quad (289)$$

Its determinant is

$$\det D_y F_{(1, 2, 1, 0)} = 2 \cdot 1 - 1 \cdot 1 = 1. \quad (290)$$

Therefore $D_y F_{(1, 2, 1, 0)}$ is invertible.

By the implicit function theorem, there exist open neighborhoods U of $(1, 2)$ and V of $(1, 0)$, and a unique C^1 function

$$h : U \longrightarrow V \quad (291)$$

such that

$$h(x_1, x_2) = \begin{pmatrix} h_1(x_1, x_2) \\ h_2(x_1, x_2) \end{pmatrix} \quad (292)$$

and

$$F(x_1, x_2, h_1(x_1, x_2), h_2(x_1, x_2)) = 0 \quad (293)$$

for all $(x_1, x_2) \in U$.

In other words, near $(x_1, x_2) = (1, 2)$, the system

$$\begin{cases} y_1^2 + y_2 = x_1, \\ y_1 + e^{y_2} = x_2 \end{cases} \quad (294)$$

implicitly defines (y_1, y_2) as a function of (x_1, x_2) .

To compute the derivative of h , we use

$$Dh_x = - (D_y F_{(x, h(x))})^{-1} D_x F_{(x, h(x))}. \quad (295)$$

In this example,

$$D_x F_{(x_1, x_2, y_1, y_2)} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (296)$$

Therefore,

$$Dh_x = (D_y F_{(x, h(x))})^{-1}. \quad (297)$$

At the point $(x_1, x_2) = (1, 2)$, we have $h(1, 2) = (1, 0)$, so

$$Dh_{(1,2)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1}. \quad (298)$$

Hence

$$Dh_{(1,2)} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}. \quad (299)$$

1.8 Extrema

Definition 1.9 (Local Extrema)

Given a function $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$,

1. $a \in D$ is a **local minimum** if there exists a neighborhood U of a such that

$$f(x) \geq f(a) \text{ for every } x \in U \quad (300)$$

2. $a \in D$ is a *local maximum* if there exists a neighborhood U of x such that

$$f(x) \leq f(a) \text{ for every } x \in U \quad (301)$$

Theorem 1.16 (1st Derivative Test)

If a is a local extremum of a differentiable function f , then $Df_a = 0$.

Note that even though the converse of this theorem is not true, we can use the contrapositive to determine that every point that has a nonzero derivative cannot be a extremum.

Example 1.22 (Function with an Infinite Number of Critical Points)

A function may also have an infinite amount of critical points (e.g. if they lie in a circle).

Unfortunately, determining whether a point a is a local extremum requires us to define an open neighborhood around a . This means that we can only determine local extrema within open sets in $\mathbb{R}^n$. Therefore, we must modify our procedure when looking for extrema on functions defined over closed bounded sets. We now describe a method of computing the global extrema. Let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a multivariable function defined on a closed and bounded set $E = E^\circ \cup \partial E$, where the E° is the interior and ∂E is the boundary of E . To find the global extrema on E , we find all local extrema of

1. f defined over the interior of E° , which is open, by locating all points where $Df = 0$, and
2. f defined over ∂E .

In the second point, we could reparamaterize ∂E cleverly by another function $\phi : \mathbb{R}^k \rightarrow \mathbb{R}^n$ (e.g. a path function for 1-dimensional manifolds ∂E) where the image coincides with ∂E , but this can be tedious. A more flexible method is to introduce a system of equality constraints of the form

$$g : \mathbb{R}^n \rightarrow \mathbb{R}^c, \quad g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix} = 0 \quad (302)$$

which really just represents a level set of g at 0, i.e. a set described by an implicit representation.¹

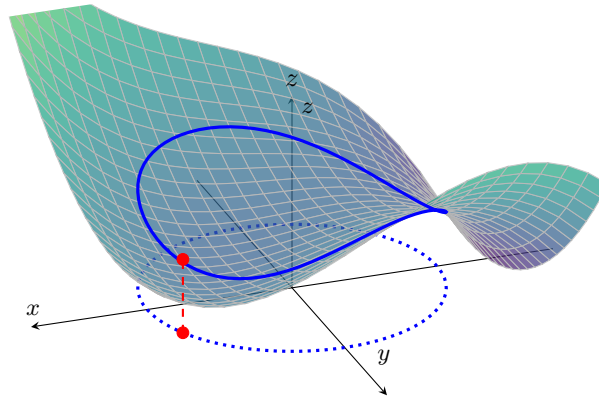


Figure 11: An example of a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ constrained to the unit circle, where $g(x, y) = x^2 + y^2 - 1 = 0$. The blue curve in the xy -plane is the constraint set, and the upper blue curve is its image on the graph of f .

To solve this constraint problem, we use the method of Lagrange multipliers. The basic idea is to convert a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

Theorem 1.17 (Lagrange Multipliers Theorem)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^c$ be C^1 functions. If

1. a is a local maximum of f restricted to $M = \{x \in \mathbb{R}^n : g(x) = 0\}$, and
2. $\text{rank}(Dg_a) = c$,

Then there exists a unique row vector $\lambda \in \mathbb{R}^{1 \times c}$ —called the **Lagrange multipliers**—such that

$$Df_a = \lambda Dg_a \quad (303)$$

Proof. Let

$$M = \{x \in \mathbb{R}^n : g(x) = 0\}.$$

Since $\text{rank}(Dg_a) = c$, the implicit function theorem tells us that near a , the constraint set M is a smooth surface of dimension $n - c$. Its tangent space at a is

$$T_a M = \ker(Dg_a).$$

Since a is a local maximum of f restricted to M , the derivative of f must vanish in every tangent direction to M . Therefore, for every $v \in T_a M$,

$$Df_a v = 0.$$

But $T_a M = \ker(Dg_a)$, so

$$v \in \ker(Dg_a) \implies Df_a v = 0.$$

Hence

$$\ker(Dg_a) \subseteq \ker(Df_a).$$

¹In physics, these types of "well-behaved" constraints are known as *holonomic constraints*.

Now we use a linear algebra fact: if a row vector annihilates $\ker(Dg_a)$, then it lies in the row space of Dg_a . Therefore Df_a lies in the row space of Dg_a . Thus there exists a row vector $\lambda \in \mathbb{R}^{1 \times c}$ such that

$$Df_a = \lambda Dg_a.$$

Finally, since $\text{rank}(Dg_a) = c$, the rows of Dg_a are linearly independent. Therefore the representation of Df_a as a linear combination of those rows is unique. Hence λ is unique.

By transposing Equation 303, we can get a matrix realization of

$$\begin{pmatrix} \frac{\partial f}{\partial x_1}(a) \\ \vdots \\ \frac{\partial f}{\partial x_n}(a) \end{pmatrix} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) & \cdots & \frac{\partial g_c}{\partial x_1}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n}(a) & \cdots & \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_c \end{pmatrix} \quad (304)$$

$$= \lambda_1 \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_1}{\partial x_n}(a) \end{pmatrix} + \lambda_2 \begin{pmatrix} \frac{\partial g_2}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_2}{\partial x_n}(a) \end{pmatrix} + \cdots + \lambda_c \begin{pmatrix} \frac{\partial g_c}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \quad (305)$$

This equation tells us that at any critical points a of f evaluated under the equality constraints, the total derivative of f at a can be expressed as a unique linear combination of the total derivatives of the constraints $D(g_i)_a$, with the Lagrange multipliers acting as coefficients. Therefore, finding the critical points a of f constrained with g is equivalent to solving the system of $c + n$ equations for the n unknowns in x and c unknowns in λ .

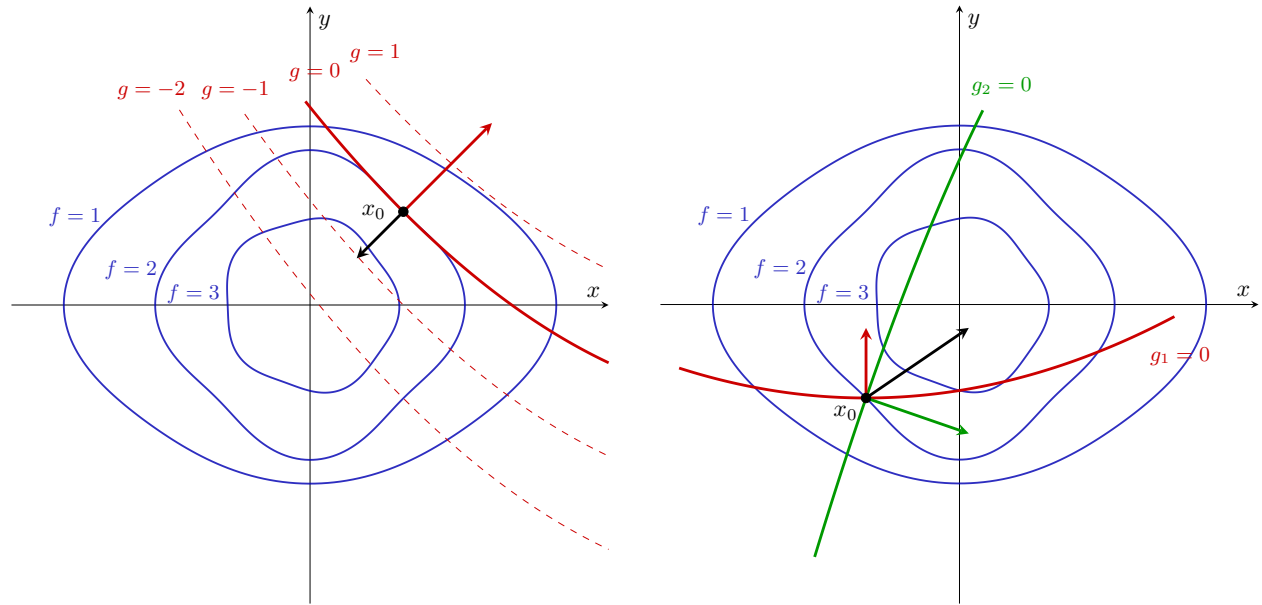
$$g(x) = 0 \iff \begin{cases} g_1(x) &= 0 \\ \dots &= 0 \\ g_c(x) &= 0 \end{cases} \quad (306)$$

$$Df_a = \lambda Dg_a \iff \begin{cases} \frac{\partial f}{\partial x_1}(a) &= \lambda_1^* \frac{\partial g_1}{\partial x_1}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_1}(a) + \cdots + \lambda_c^* \frac{\partial g_c}{\partial x_1}(a) \\ \dots &= \dots \\ \frac{\partial f}{\partial x_n}(a) &= \lambda_1^* \frac{\partial g_1}{\partial x_n}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_n}(a) + \cdots + \lambda_c^* \frac{\partial g_c}{\partial x_n}(a) \end{cases} \quad (307)$$

This relationship between the gradient of the function and gradients of the constraints rather naturally leads to a reformulation of the original problem, known as the *Lagrangian function*. That is, in order to find the maximum/minimum of f subjected to the equality constraint $g(x) = 0$, we form the Lagrangian function

$$\mathcal{L}(x, \lambda) = f(x) - \lambda^T g(x) \quad (308)$$

and find the stationary points of $\mathcal{L}$ considered as a function of $x \in E$ and the Lagrange multiplier $\lambda \in \mathbb{R}$. The main advantage to this method is that it allows the optimization to be solved without explicit parameterization in terms of the constraints.



(a) When there is a single constraint $g : \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g(x_0)$.

(b) When there are multiple constraints g_1, g_2 , $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g_1(x_0)$ and $\nabla g_2(x_0)$.

Figure 12: From the properties of the gradient, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = f(x_0)$ at point x_0 . We can see that the point where the contour line of $g(x) = 0$ tangentially touches the contours of f is the maximum. Since it intersects it tangentially, the gradient vector at that point $\nabla g(x_0)$ is parallel to $\nabla f(x_0)$.

From the properties of the gradient introduced before, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = c$ at the point x_0 . But this level set $f(x) = c$ actually intersects the level set determined by $g(x) = c$ at the point x_0 and is indistinguishable from each other at x_0 . This means that $\nabla g(x_0)$ is normal the level set of $g(x) = c$ at $x_0 \iff$ it is normal to the level set of $f(x) = c$ at x_0 . But $\nabla f(x_0)$ is also normal at that point, so $\nabla f(x_0)$ must be parallel to $\nabla g(x_0)$.